33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F13-SR-002
415V BALL MILL SWITCHROOM

F13-TR-003
TRANSFORMER ROOM

F13-TR-003
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
4000/ 1A
2.5VA
5P20

F13-SW-003
BALL MILL 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
5P20
00F-01
4000/ 5A
30VA
C10.5
00F-08
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
ACB
4000A
AFD
50G
2A
24VDC CONTROL POWER SUPPLY
NOTE 7
50
51
50G
51G
125A
160A
SD
NOTE 6

Approved [X]
Approved as Noted [ ]
Not Approved [ ]
For Information Only [ ]
Signed [redacted]
Date 09/10/2025
ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 4000A 65kA for 1sec
INC
6F-04 63A 70A
6F-01 100A 90A
5R-01 20A 70A
3F-01 20A 70A
1F-09 32A 70A
1F-13 32A 70A
7F-01 800A 400A
1F-17 32A 70A
1F-21 32A 70A
2F-01 32A 70A
2F-05 32A 70A
2F-09 32A 70A
2A 32A
EOL

CONTINUED ON SHT 2
2000-F13-SLD-EL-10902

FEEDER TO F13-SW-103
2000-F13-SCM-EL-10952

E-SHUT
F13SR002-LCS
NOTE 12

CP	20kW	F13-BLC-402 BALL MILL BALL CHARGER 2000-F13-SCM-EL-10909
CP	29kW	F13CU408-CP BALL MILL VSD COOLING SYSTEM CONTROL PANEL 2000-F13-SCM-EL-10912
CP	6kW	F13CU409-CP CYCLONE FEED PUMP WATER COOLER UNIT CONTROL PANEL 2000-F13-SCM-EL-10942
M 3~	0.37kW	F13-FN-426 BALL MILL MOTOR LUBE SKID COOLER FAN 2000-F13-SCM-EL-10913
M 3~	7.5kW	F13-FN-435 BALL MILL MOTOR 1 FAN 1 2000-F13-SCM-EL-10914
M 3~	7.5kW	F13-FN-436 BALL MILL MOTOR 1 FAN 2 2000-F13-SCM-EL-10915
M 3~	7.5kW	F13-FN-437 BALL MILL MOTOR 1 FAN 3 2000-F13-SCM-EL-10916
M 3~	7.5kW	F13-FN-438 BALL MILL MOTOR 1 FAN 4 2000-F13-SCM-EL-10917
M 3~	7.5kW	F13-FN-439 BALL MILL MOTOR 2 FAN 1 2000-F13-SCM-EL-10918
M 3~	7.5kW	F13-FN-440 BALL MILL MOTOR 2 FAN 2 2000-F13-SCM-EL-10919
M 3~	7.5kW	F13-FN-441 BALL MILL MOTOR 2 FAN 3 2000-F13-SCM-EL-10920

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-003 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10002.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA — ST TRIP SET POINT; xxxA — CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES
2000-F13-SCM-EL-10903	F13-SW-003 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F13-SCM-EL-10902	F13-SW-003 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F13-SCM-EL-10901	F13-SW-003 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F13-DRG-EL-10102	F13-SR-002 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F13-BLK-IC-10002	F13-SR-002 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F13-SLD-EL-10902	F13-SW-003 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	28AUG25	FAT REDLINES INCORPORATED					
0	29MAY24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION - BALL MILL
F13-SW-003 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F13-SLD-EL-10901	1

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